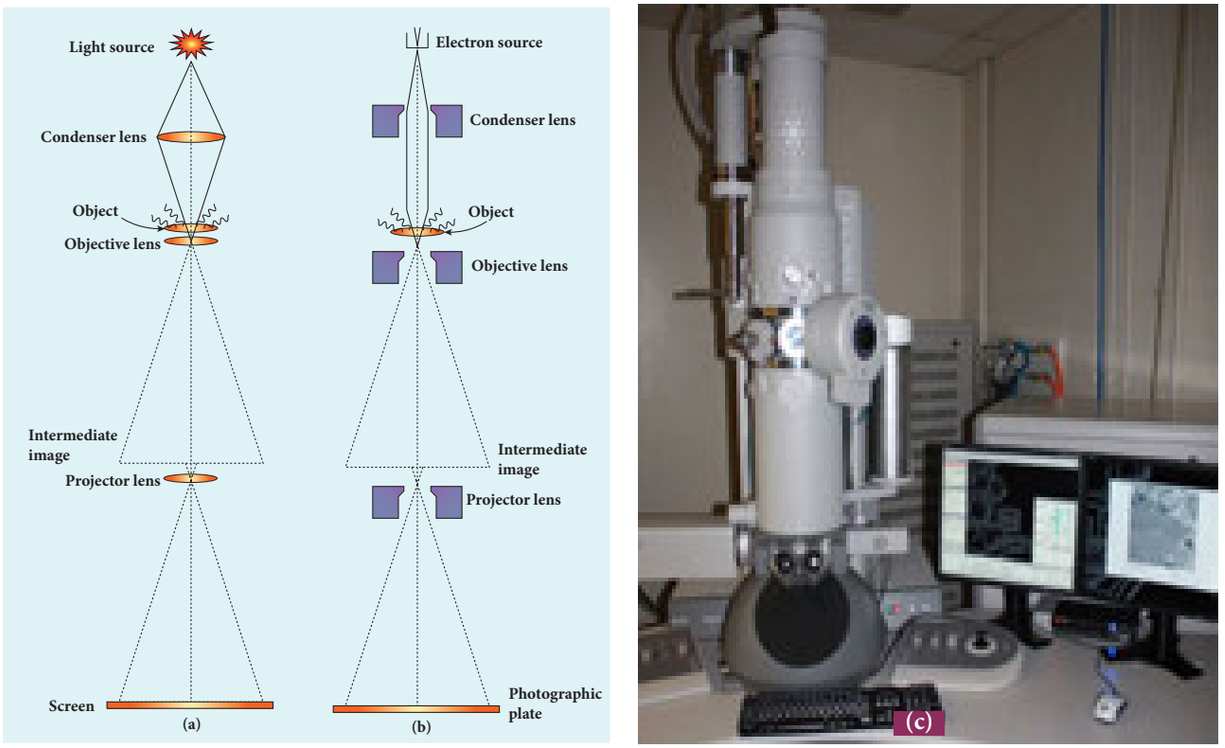


Figure 8.19 (a) Optical microscope (b) Electron microscope (c) Photograph of electron microscope

optical microscope. Electron microscopes giving magnification more than 2,00,000 times are common in research laboratories.

Working

The construction and working of an electron microscope is similar to that of an optical microscope except that in electron microscope focussing of electron beam is done by the electrostatic or magnetic lenses. The electron beam passing across a suitably arranged either electric or magnetic fields undergoes divergence or convergence thereby focussing of the beam is done (Figure 8.19).

The electrons emitted from the source are accelerated by high potentials. The beam is made parallel by magnetic condenser lens. When the beam passes through the sample whose magnified image is needed, the beam carries the image of the sample.

With the help of magnetic objective lens and magnetic projector lens system, the magnified image is obtained on the screen. These electron microscopes are being used in almost all branches of science.

EXAMPLE 8.6

Calculate the momentum and the de Broglie wavelength in the following cases:

i) an electron with kinetic energy 2 eV.
ii) a bullet of 50 g fired from rifle with a speed of 200 m/s
iii) a 4000 kg car moving along the highways at 50 m/s

Hence show that the wave nature of matter is important at the atomic level but is not really relevant at macroscopic level.

Solution:

i) Momentum of the electron is

$$p = \sqrt{2mK} = \sqrt{2\times 9.1\times 10^{-31}\times 2\times 1.6\times 10^{-19}}$$
$$= 7.63\times 10^{-25}\,\text{kg m s}^{-1}$$

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